

Ibn Tofail University

Rigid body mechanics

Problem Set I

Exercise 1:

Let $(A, \vec{V})$ be a bound vector defined in the basis $(\vec{i}, \vec{j}, \vec{k})$ by: $A(-2, -1, 2)$ and $\vec{V} = \vec{i} + 2\vec{j} - \vec{k}$.

1. Calculate the moment of the vector $\vec{V}$ at the point with coordinates $(2, 1, 1)$.
2. Calculate the moment of $(A, \vec{V})$ with respect to the axis Δ passing through B and parallel to the vector $\vec{U} = 2\vec{i} - 3\vec{k}$.

Correction

1. Moment at point $P(2, 1, 1)$

Given:

$$A(-2, -1, 2), \quad \vec{V} = \vec{i} + 2\vec{j} - \vec{k}, \quad P(2, 1, 1)$$

$$\vec{PA} = A - P = (-4, -2, 1)$$

$$\vec{M}_P = \vec{PA} \times \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -4 & -2 & 1 \\ 1 & 2 & -1 \end{vmatrix}$$

$$= \vec{i}[(-2)(-1) - (1)(2)] - \vec{j}[(-4)(-1) - (1)(1)] + \vec{k}[(-4)(2) - (-2)(1)]$$

$$= \vec{i}(2 - 2) - \vec{j}(4 - 1) + \vec{k}(-8 + 2)$$

$$= 0\vec{i} - 3\vec{j} - 6\vec{k}$$

$$\boxed{-3\vec{j} - 6\vec{k}}$$

2. Moment about axis Δ through B parallel to $\vec{U}$

Assuming $B = O(0, 0, 0)$ (since B not specified):

$$\vec{BA} = A - B = (-2, -1, 2)$$

$$\vec{M}_B = \vec{BA} \times \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & -1 & 2 \\ 1 & 2 & -1 \end{vmatrix}$$

$$= \vec{i}[(-1)(-1) - (2)(2)] - \vec{j}[(-2)(-1) - (2)(1)] + \vec{k}[(-2)(2) - (-1)(1)]$$

$$= \vec{i}(1 - 4) - \vec{j}(2 - 2) + \vec{k}(-4 + 1)$$

$$= -3\vec{i} + 0\vec{j} - 3\vec{k}$$

Given $\vec{U} = 2\vec{i} - 3\vec{k}$, scalar projection:

$$M_{\Delta} = \frac{\vec{M}_B \cdot \vec{U}}{\|\vec{U}\|}$$

but vector moment along axis:

$$\vec{M}_{\Delta} = \left(\frac{\vec{M}_B \cdot \vec{U}}{\|\vec{U}\|^2} \right) \vec{U}$$

$$\vec{M}_B \cdot \vec{U} = (-3)(2) + (0)(0) + (-3)(-3) = -6 + 9 = 3$$

$$\|\vec{U}\|^2 = 2^2 + 0^2 + (-3)^2 = 13$$

$$\vec{M}_{\Delta} = \frac{3}{13}(2\vec{i} - 3\vec{k}) = \frac{6}{13}\vec{i} - \frac{9}{13}\vec{k}$$

$$\boxed{\frac{6}{13}\vec{i} - \frac{9}{13}\vec{k}}$$

Exercise 2:

We are given two torsors T_1 and T_2 . Find the set of points (if it exists) in space where their moments are collinear.

Correction

Let torsors T_1 and T_2 be defined by:

$$T_1 : \text{Resultant } \vec{R}_1, \quad \text{Moment at } O : \vec{M}_1(O)$$

$$T_2 : \text{Resultant } \vec{R}_2, \quad \text{Moment at } O : \vec{M}_2(O)$$

At point Q with position vector $\vec{r} = O\vec{Q}$:

$$\vec{M}_1(Q) = \vec{M}_1(O) + \vec{R}_1 \times \vec{r}$$

$$\vec{M}_2(Q) = \vec{M}_2(O) + \vec{R}_2 \times \vec{r}$$

Collinearity condition:

$$\vec{M}_1(Q) \times \vec{M}_2(Q) = \vec{0}$$

Substitute:

$$[\vec{M}_1(O) + \vec{R}_1 \times \vec{r}] \times [\vec{M}_2(O) + \vec{R}_2 \times \vec{r}] = \vec{0}$$

Let $\vec{A} = \vec{M}_1(O)$, $\vec{B} = \vec{M}_2(O)$, $\vec{u} = \vec{R}_1$, $\vec{v} = \vec{R}_2$:

$$(\vec{A} + \vec{u} \times \vec{r}) \times (\vec{B} + \vec{v} \times \vec{r}) = \vec{0}$$

Expanding:

$$\vec{A} \times \vec{B} + \vec{A} \times (\vec{v} \times \vec{r}) + (\vec{u} \times \vec{r}) \times \vec{B} + (\vec{u} \times \vec{r}) \times (\vec{v} \times \vec{r}) = \vec{0}$$

Using vector identities:

$$\begin{aligned}\vec{A} \times (\vec{v} \times \vec{r}) &= (\vec{A} \cdot \vec{r})\vec{v} - (\vec{A} \cdot \vec{v})\vec{r} \\ (\vec{u} \times \vec{r}) \times \vec{B} &= -(\vec{B} \cdot \vec{r})\vec{u} + (\vec{B} \cdot \vec{u})\vec{r} \\ (\vec{u} \times \vec{r}) \times (\vec{v} \times \vec{r}) &= -[(\vec{u} \times \vec{r}) \cdot \vec{v}]\vec{r} = -[\vec{r} \cdot (\vec{v} \times \vec{u})]\vec{r}\end{aligned}$$

Let $\vec{w} = \vec{v} \times \vec{u} = -\vec{u} \times \vec{v}$:

$$\vec{A} \times \vec{B} + (\vec{A} \cdot \vec{r})\vec{v} - (\vec{A} \cdot \vec{v})\vec{r} - (\vec{B} \cdot \vec{r})\vec{u} + (\vec{B} \cdot \vec{u})\vec{r} - (\vec{r} \cdot \vec{w})\vec{r} = \vec{0}$$

This is a **vector quadratic equation** in $\vec{r}$. The solution set is generally a quadric surface, which may degenerate to a plane, line, point, or be empty depending on the specific torsors.

The set of points forms a quadric surface, possibly degenerate

Exercise 3:

Let E be a three-dimensional vector space and S a linear application from E to E which associates to any vector $\vec{U}$ of E its image $\vec{U}' = S(\vec{U})$. We are interested in the case where S is an antisymmetric application defined by: $S(\vec{U}) = L \cdot \vec{U}$.

Find the matrix L associated with the antisymmetric application S by taking s_1, s_2 , and s_3 as components of the vector $\vec{S}$.

Correction

Let S be an antisymmetric linear application:

$$S(\vec{U}) = L \cdot \vec{U}$$

with S antisymmetric: $S^T = -S$.

In 3D, any antisymmetric linear operator corresponds to a cross product with some fixed vector $\vec{s} = (s_1, s_2, s_3)$:

$$S(\vec{U}) = \vec{s} \times \vec{U}$$

Let's find matrix L such that $\vec{s} \times \vec{U} = L\vec{U}$.

For $\vec{U} = (u_1, u_2, u_3)$:

$$\vec{s} \times \vec{U} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ s_1 & s_2 & s_3 \\ u_1 & u_2 & u_3 \end{vmatrix} = (s_2u_3 - s_3u_2, s_3u_1 - s_1u_3, s_1u_2 - s_2u_1)$$

This equals:

$$\begin{pmatrix} 0 & -s_3 & s_2 \\ s_3 & 0 & -s_1 \\ -s_2 & s_1 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

Thus the matrix L associated with the antisymmetric application S is:

$$L = \begin{pmatrix} 0 & -s_3 & s_2 \\ s_3 & 0 & -s_1 \\ -s_2 & s_1 & 0 \end{pmatrix}$$

where s_1, s_2, s_3 are the components of the vector $\vec{S}$.

Exercise 4:

To every point $P(x, y, z)$ of affine space, we associate the family of vector fields $\vec{V}_t(P)$ defined by:

$$\vec{V}_t(P) = (3y - tz + 1, -3x + 2tz, tx + t^2y - 4/3); \quad t \in \mathbb{R}.$$

1. Show that the only equiprojective fields are obtained for $t = 0$ and $t = 2$.
2. Determine for each case the associated torsors by their reduction elements at point $O(0, 0, 0)$.

Correction

Given:

$$\vec{V}_t(P) = (3y - tz + 1, -3x + 2tz, tx + t^2y - \frac{4}{3}); \quad t \in \mathbb{R}$$

Part 1: Equiprojective fields

A vector field $\vec{V}$ is equiprojective if:

$$\vec{V}(P) \cdot \vec{PQ} = \vec{V}(Q) \cdot \vec{PQ} \quad \forall P, Q$$

Equivalently: $\vec{V}$ is affine and its linear part is antisymmetric.

Let $\vec{V}(x, y, z) = (V_x, V_y, V_z)$ with:

$$V_x = 3y - tz + 1$$

$$V_y = -3x + 2tz$$

$$V_z = tx + t^2y - \frac{4}{3}$$

The linear part matrix is:

$$J = \begin{pmatrix} 0 & 3 & -t \\ -3 & 0 & 2t \\ t & t^2 & 0 \end{pmatrix}$$

For equiprojectivity, J must be antisymmetric: $J^T = -J$.

Check antisymmetry:

- $J_{12} = 3, -J_{21} = 3$
- $J_{13} = -t, -J_{31} = -t$
- $J_{23} = 2t, -J_{32} = -t^2 \implies 2t = -t^2$

From $2t = -t^2$: $t^2 + 2t = 0 \implies t(t+2) = 0 \implies t = 0$ or $t = -2$
 Thus equiprojective fields occur for:

$$t = 0 \quad \text{and} \quad t = -2$$

Part 2: Associated torsors at $O(0,0,0)$

Case 1: $t = 0$

$$\vec{V}_0(P) = (3y + 1, -3x, -\frac{4}{3})$$

Resultant $\vec{R} = \text{constant part} = (1, 0, -\frac{4}{3})$

Moment at O : $\vec{M}_O = \vec{V}_0(O) = (1, 0, -\frac{4}{3})$

Torsor at O : $\{\vec{R} = (1, 0, -\frac{4}{3}), \vec{M}_O = (1, 0, -\frac{4}{3})\}$

Case 2: $t = -2$

$$\vec{V}_{-2}(P) = (3y + 2z + 1, -3x - 4z, -2x + 4y - \frac{4}{3})$$

Resultant $\vec{R} = \text{constant part} = (1, 0, -\frac{4}{3})$

Moment at O : $\vec{M}_O = \vec{V}_{-2}(O) = (1, 0, -\frac{4}{3})$

Torsor at O : $\{\vec{R} = (1, 0, -\frac{4}{3}), \vec{M}_O = (1, 0, -\frac{4}{3})\}$

$$\text{For both } t = 0 \text{ and } t = -2: \vec{R} = (1, 0, -\frac{4}{3}), \vec{M}_O = (1, 0, -\frac{4}{3})$$

Exercise 5:

We are given two sliders $(A, \vec{U})$ and $(B, \vec{V})$ such that: $A(1, 1, \alpha)$, $B(0, 2, 0)$, $\vec{U}(0, 0, \alpha)$ and $\vec{V}(\beta, 3, 0)$ where α and β are real numbers. Let the torsor $T = (A, \vec{U}) + (B, \vec{V})$.

1. Give the reduction elements of T at point O .
2. What is the necessary and sufficient condition for T to be a slider?
3. Determine the support of T .

Correction

Given sliders:

$$(A, \vec{U}) : A(1, 1, \alpha), \vec{U}(0, 0, \alpha)$$

$$(B, \vec{V}) : B(0, 2, 0), \vec{V}(\beta, 3, 0)$$

Torsor: $T = (A, \vec{U}) + (B, \vec{V})$

Part 1: Reduction elements at point $O(0,0,0)$

Resultant vector:

$$\vec{R} = \vec{U} + \vec{V} = (0 + \beta, 0 + 3, \alpha + 0) = (\beta, 3, \alpha)$$

Moment at O :

$$\vec{M}_O = \vec{OA} \times \vec{U} + \vec{OB} \times \vec{V}$$

$$\vec{OA} \times \vec{U} = (1, 1, \alpha) \times (0, 0, \alpha) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & \alpha \\ 0 & 0 & \alpha \end{vmatrix} = (\alpha, -\alpha, 0)$$

$$\vec{OB} \times \vec{V} = (0, 2, 0) \times (\beta, 3, 0) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 2 & 0 \\ \beta & 3 & 0 \end{vmatrix} = (0, 0, -2\beta)$$

$$\vec{M}_O = (\alpha, -\alpha, 0) + (0, 0, -2\beta) = (\alpha, -\alpha, -2\beta)$$

Reduction at O :

$$\boxed{\vec{R} = (\beta, 3, \alpha), \quad \vec{M}_O = (\alpha, -\alpha, -2\beta)}$$

Part 2: Condition for T to be a slider

A torsor is a slider if $\vec{R} \cdot \vec{M}_O = 0$ (invariant scalar is zero).

$$\vec{R} \cdot \vec{M}_O = (\beta, 3, \alpha) \cdot (\alpha, -\alpha, -2\beta) = \beta\alpha - 3\alpha - 2\alpha\beta = -3\alpha$$

Thus:

$$\vec{R} \cdot \vec{M}_O = 0 \iff -3\alpha = 0 \iff \alpha = 0$$

$$\boxed{\alpha = 0}$$

No condition on β .

Part 3: Support of T

The central axis (support) is given by:

$$\vec{M}_P = \vec{M}_O + \vec{R} \times \vec{OP} = \lambda \vec{R}$$

Let $P(x, y, z)$, $\vec{OP} = (x, y, z)$:

$$\vec{R} \times \vec{OP} = (\beta, 3, \alpha) \times (x, y, z) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \beta & 3 & \alpha \\ x & y & z \end{vmatrix} = (3z - \alpha y, \alpha x - \beta z, \beta y - 3x)$$

So:

$$\vec{M}_P = (\alpha + 3z - \alpha y, -\alpha + \alpha x - \beta z, -2\beta + \beta y - 3x)$$

Set $\vec{M}_P = \lambda \vec{R} = (\lambda\beta, 3\lambda, \lambda\alpha)$:

System:

$$\alpha + 3z - \alpha y = \lambda\beta \quad (1)$$

$$-\alpha + \alpha x - \beta z = 3\lambda \quad (2)$$

$$-2\beta + \beta y - 3x = \lambda\alpha \quad (3)$$

Eliminating λ gives the central axis equations. When $\alpha = 0$ (slider case), system simplifies.

For general case, the support is a line determined by solving this system.

$$\boxed{\text{Support is the central axis given by } \vec{M}_P = \lambda \vec{R}}$$

Exercise 6:

Let three sliders $(A_i, \vec{V}_i)$; $i = 1, 2, 3$ such that: $A_1(0, 0, a)$, $A_2(a, 0, 0)$ and $A_3(0, a, 0)$; $\vec{V}_1(2a, 0, 0)$, $\vec{V}_2(0, 3a, 0)$ and $\vec{V}_3(0, 0, \lambda a)$. We consider the torsor $L = \sum_{i=1}^3 (A_i, \vec{V}_i)$. Determine the reduction elements of L at an arbitrary point P as well as its central axis.

Correction

Given sliders:

$$(A_1, \vec{V}_1) : A_1(0, 0, a), \vec{V}_1(2a, 0, 0)$$

$$(A_2, \vec{V}_2) : A_2(a, 0, 0), \vec{V}_2(0, 3a, 0)$$

$$(A_3, \vec{V}_3) : A_3(0, a, 0), \vec{V}_3(0, 0, \lambda a)$$

Torsor: $L = \sum_{i=1}^3 (A_i, \vec{V}_i)$

Reduction elements at arbitrary point $P(x, y, z)$

Resultant vector:

$$\vec{R} = \vec{V}_1 + \vec{V}_2 + \vec{V}_3 = (2a, 0, 0) + (0, 3a, 0) + (0, 0, \lambda a) = (2a, 3a, \lambda a)$$

Moment at O :

$$\begin{aligned} \vec{M}_O &= \vec{OA}_1 \times \vec{V}_1 + \vec{OA}_2 \times \vec{V}_2 + \vec{OA}_3 \times \vec{V}_3 \\ &= (0, 0, a) \times (2a, 0, 0) + (a, 0, 0) \times (0, 3a, 0) + (0, a, 0) \times (0, 0, \lambda a) \\ &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & a \\ 2a & 0 & 0 \end{vmatrix} + \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a & 0 & 0 \\ 0 & 3a & 0 \end{vmatrix} + \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & a & 0 \\ 0 & 0 & \lambda a \end{vmatrix} \\ &= (0, 2a^2, 0) + (0, 0, 3a^2) + (\lambda a^2, 0, 0) \\ &= (\lambda a^2, 2a^2, 3a^2) \end{aligned}$$

Moment at $P(x, y, z)$:

$$\begin{aligned} \vec{M}_P &= \vec{M}_O + \vec{R} \times \vec{OP} \\ \vec{R} \times \vec{OP} &= (2a, 3a, \lambda a) \times (x, y, z) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2a & 3a & \lambda a \\ x & y & z \end{vmatrix} = a(3z - \lambda y, \lambda x - 2z, 2y - 3x) \end{aligned}$$

Thus:

$$\vec{M}_P = (\lambda a^2 + a(3z - \lambda y), 2a^2 + a(\lambda x - 2z), 3a^2 + a(2y - 3x))$$

Central axis

Central axis defined by $\vec{M}_P = \mu \vec{R}$:

$$(\lambda a^2 + a(3z - \lambda y), 2a^2 + a(\lambda x - 2z), 3a^2 + a(2y - 3x)) = \mu(2a, 3a, \lambda a)$$

Divide by a ($a \neq 0$):

$$\lambda a + 3z - \lambda y = 2\mu \quad (1)$$

$$2a + \lambda x - 2z = 3\mu \quad (2)$$

$$3a + 2y - 3x = \lambda\mu \quad (3)$$

From (1) and (2), eliminate μ :

$$\mu = \frac{\lambda a + 3z - \lambda y}{2} = \frac{2a + \lambda x - 2z}{3}$$

Multiply by 6:

$$\begin{aligned} 3\lambda a + 9z - 3\lambda y &= 4a + 2\lambda x - 4z \\ 3\lambda a - 4a - 3\lambda y - 2\lambda x + 9z + 4z &= 0 \\ a(3\lambda - 4) - \lambda(3y + 2x) + 13z &= 0 \quad (4) \end{aligned}$$

From (1) and (3), substitute μ from (1) into (3):

$$3a + 2y - 3x = \lambda \cdot \frac{\lambda a + 3z - \lambda y}{2}$$

Multiply by 2:

$$\begin{aligned} 6a + 4y - 6x &= \lambda^2 a + 3\lambda z - \lambda^2 y \\ 6a - \lambda^2 a + 4y + \lambda^2 y - 6x - 3\lambda z &= 0 \\ a(6 - \lambda^2) + y(4 + \lambda^2) - 6x - 3\lambda z &= 0 \quad (5) \end{aligned}$$

Equations (4) and (5) define the central axis.

$$\vec{R} = (2a, 3a, \lambda a), \quad \vec{M}_P = a(\lambda a + 3z - \lambda y, 2a + \lambda x - 2z, 3a + 2y - 3x)$$

Central axis: equations (4) and (5)

Exercise 7:

Consider, in a Cartesian frame $R(O, X, Y, Z)$ with basis $(\vec{i}, \vec{j}, \vec{k})$, the following system of three bound vectors defined by:

$$\begin{aligned} A_1(1, 0, 0) \quad \vec{V}_1 &= a\vec{i} + b\vec{j} + \vec{k} \\ A_2(0, 1, 0) \quad \vec{V}_2 &= 2b\vec{i} - 2a\vec{j} - 3\vec{k} \\ A_3(0, 0, 1) \quad \vec{V}_3 &= -8\vec{i} + \vec{j} + c\vec{k} \end{aligned}$$

where a , b , and c are real constants.

For what values of a , b , and c is the system of three bound vectors equivalent to a couple? Determine then its moment.

Correction

Given three bound vectors:

$$\begin{aligned} A_1(1, 0, 0) \quad \vec{V}_1 &= a\vec{i} + b\vec{j} + \vec{k} \\ A_2(0, 1, 0) \quad \vec{V}_2 &= 2b\vec{i} - 2a\vec{j} - 3\vec{k} \\ A_3(0, 0, 1) \quad \vec{V}_3 &= -8\vec{i} + \vec{j} + c\vec{k} \end{aligned}$$

Condition for equivalence to a couple

A system is equivalent to a couple if the resultant vector is zero:

$$\vec{R} = \vec{V}_1 + \vec{V}_2 + \vec{V}_3 = \vec{0}$$

Compute resultant:

$$\begin{aligned}\vec{R} &= (a + 2b - 8)\vec{i} + (b - 2a + 1)\vec{j} + (1 - 3 + c)\vec{k} \\ &= (a + 2b - 8)\vec{i} + (b - 2a + 1)\vec{j} + (c - 2)\vec{k}\end{aligned}$$

Set $\vec{R} = \vec{0}$:

$$a + 2b - 8 = 0 \quad (1)$$

$$-2a + b + 1 = 0 \quad (2)$$

$$c - 2 = 0 \quad (3)$$

From (3): $\boxed{c = 2}$

Solve (1) and (2): From (2): $b = 2a - 1$ Substitute in (1): $a + 2(2a - 1) - 8 = 0$
 $5a - 10 = 0 \implies a = 2$ Then $b = 2(2) - 1 = 3$

Thus:

$$\boxed{a = 2, \quad b = 3, \quad c = 2}$$

Moment of the couple

When $\vec{R} = \vec{0}$, the moment is invariant (same at all points). Compute at O :

$$\begin{aligned}\vec{M}_O &= \vec{OA}_1 \times \vec{V}_1 + \vec{OA}_2 \times \vec{V}_2 + \vec{OA}_3 \times \vec{V}_3 \\ &= (1, 0, 0) \times (2, 3, 1) + (0, 1, 0) \times (6, -4, -3) + (0, 0, 1) \times (-8, 1, 2)\end{aligned}$$

Compute each cross product:

$$(1, 0, 0) \times (2, 3, 1) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 2 & 3 & 1 \end{vmatrix} = (0, -1, 3)$$

$$(0, 1, 0) \times (6, -4, -3) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 0 \\ 6 & -4 & -3 \end{vmatrix} = (-3, 0, -6)$$

$$(0, 0, 1) \times (-8, 1, 2) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 1 \\ -8 & 1 & 2 \end{vmatrix} = (-1, -8, 0)$$

Sum:

$$\vec{M}_O = (0 - 3 - 1, -1 + 0 - 8, 3 - 6 + 0) = (-4, -9, -3)$$

$$\boxed{\vec{M} = -4\vec{i} - 9\vec{j} - 3\vec{k}}$$

Exercise 8:

We consider the vector field $\vec{W}$ which associates to every point $M(x, y, z)$ the vector $\vec{W}(M)$ with components:

$$\begin{aligned} W_x &= -1 - y - 2az \\ W_y &= 3 + x - 2az \\ W_z &= -2 + 2ax + a^2y \end{aligned}$$

where a is a real parameter.

1. For what values of a is the field $\vec{W}$ an antisymmetric field?
2. Determine, in each case, the reduction elements of the associated torsor.

Correction

Given:

$$\vec{W}(M) = (-1 - y - 2az)\vec{i} + (3 + x - 2az)\vec{j} + (-2 + 2ax + a^2y)\vec{k}$$

Part 1: Condition for antisymmetric field

A vector field $\vec{W}$ is antisymmetric if it can be written as:

$$\vec{W}(M) = \vec{W}_O + \vec{\omega} \times O\vec{M}$$

where $\vec{\omega}$ is constant (the rotation vector).

Equivalently, the Jacobian matrix J of $\vec{W}$ must be antisymmetric.

Compute partial derivatives:

$$\begin{array}{lll} \frac{\partial W_x}{\partial x} = 0, & \frac{\partial W_x}{\partial y} = -1, & \frac{\partial W_x}{\partial z} = -2a \\ \frac{\partial W_y}{\partial x} = 1, & \frac{\partial W_y}{\partial y} = 0, & \frac{\partial W_y}{\partial z} = -2a \\ \frac{\partial W_z}{\partial x} = 2a, & \frac{\partial W_z}{\partial y} = a^2, & \frac{\partial W_z}{\partial z} = 0 \end{array}$$

Jacobian matrix:

$$J = \begin{pmatrix} 0 & -1 & -2a \\ 1 & 0 & -2a \\ 2a & a^2 & 0 \end{pmatrix}$$

For antisymmetry: $J^T = -J$

Check conditions:

- $J_{12} = -1, -J_{21} = -1$
- $J_{13} = -2a, -J_{31} = -2a \implies -2a = -2a$
- $J_{23} = -2a, -J_{32} = -a^2 \implies -2a = -a^2$

From $-2a = -a^2$: $a^2 - 2a = 0 \implies a(a - 2) = 0$

Thus:

$$\boxed{a = 0 \quad \text{or} \quad a = 2}$$

Part 2: Reduction elements of associated torsor

Case 1: $a = 0$

$$\vec{W}(M) = (-1 - y)\vec{i} + (3 + x)\vec{j} + (-2)\vec{k}$$

Resultant $\vec{R} = \vec{\omega}$ from $\vec{W}(M) = \vec{W}_O + \vec{\omega} \times O\vec{M}$

Identify constant part: $\vec{W}_O = (-1, 3, -2)$

Identify rotation vector from antisymmetric part:

$$J = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \vec{\omega} = (0, 0, 1)$$

Check: $\vec{\omega} \times O\vec{M} = (0, 0, 1) \times (x, y, z) = (-y, x, 0)$

Thus: $\vec{W}(M) = (-1, 3, -2) + (-y, x, 0) = (-1 - y, 3 + x, -2)$

Reduction at O :

$$\vec{R} = \vec{\omega} = (0, 0, 1), \quad \vec{M}_O = \vec{W}_O = (-1, 3, -2)$$

Case 2: $a = 2$

$$\vec{W}(M) = (-1 - y - 4z)\vec{i} + (3 + x - 4z)\vec{j} + (-2 + 4x + 4y)\vec{k}$$

Constant part: $\vec{W}_O = (-1, 3, -2)$

Jacobian:

$$J = \begin{pmatrix} 0 & -1 & -4 \\ 1 & 0 & -4 \\ 4 & 4 & 0 \end{pmatrix} \Rightarrow \vec{\omega} = (4, -4, 1)$$

Check: $\vec{\omega} \times O\vec{M} = (4, -4, 1) \times (x, y, z) = (-4z - 4y, 4z - x, 4y + 4x)$

Thus: $\vec{W}(M) = (-1, 3, -2) + (-4z - 4y, 4z - x, 4y + 4x) = (-1 - y - 4z, 3 + x - 4z, -2 + 4x + 4y)$

Reduction at O :

$$\vec{R} = \vec{\omega} = (4, -4, 1), \quad \vec{M}_O = \vec{W}_O = (-1, 3, -2)$$

$$\boxed{a = 0 : \vec{R} = (0, 0, 1), \vec{M}_O = (-1, 3, -2) \quad a = 2 : \vec{R} = (4, -4, 1), \vec{M}_O = (-1, 3, -2)}$$